

Task 7 – First Chart Story: Turning Numbers into a Narrative

World Happiness Report Data Story | Veda Technologies – Data Analytics Internship

1. Project Overview

This project focuses on turning a small set of meaningful data points into a clear visual story. Using the World Happiness Report dataset, the analysis explores reported life evaluation and its relationship with selected social and economic indicators. Python and Matplotlib are used to create four simple visualizations, with each chart answering a specific question.

2. Objective

The main objective is to learn how to select an appropriate chart type for a business or analytical question, present information clearly, and connect multiple visualizations into one understandable narrative.

3. Tools and Technologies

Tool	Purpose
Python	Data analysis and visualization workflow
Pandas	Loading, cleaning, grouping and analyzing data
Matplotlib	Creating charts and visual storytelling
Google Colab	Running the Python notebook in a cloud environment
Excel Dataset	World Happiness Report data source

4. Dataset

The World Happiness Report dataset is used for this analysis. The dataset contains country-level measurements related to life evaluation and several factors associated with well-being, including economic well-being, social support, and freedom to make life choices.

5. Visualizations

Chart	Question Answered	Chart Type
1. Top 10 Countries by Average Life Evaluation	Which countries have the highest reported life evaluations?	Horizontal Bar Chart
2. Economic Well-being vs Life Evaluation	Is higher economic well-being associated with higher life evaluation?	Scatter Plot
3. Social Support vs Life Evaluation	Do countries with stronger social support tend to report higher life evaluation?	Scatter Plot
4. Freedom vs Life Evaluation	Is freedom to make life choices associated with reported life evaluation?	Scatter Plot

6. Chart-wise Takeaways

Chart 1 – Top 10 Countries

The chart highlights the countries with the highest average reported life evaluations in the selected dataset.

Chart 2 – GDP vs Life Evaluation

The scatter plot helps examine whether countries with higher economic well-being also tend to report higher life evaluation.

Chart 3 – Social Support vs Life Evaluation

The visualization helps examine whether stronger social support is associated with higher reported life evaluation.

Chart 4 – Freedom vs Life Evaluation

The chart explores the relationship between perceived freedom to make life choices and reported life evaluation.

7. Data Story

Happiness is influenced by more than a single factor. The analysis first identifies countries with higher reported life evaluations and then explores several factors that may be associated with differences in well-being. Economic well-being is examined through GDP per capita, followed by social support and freedom to make life choices. Together, these visualizations provide a broader perspective on how different social and economic indicators relate to reported life evaluation across countries.

8. Analysis Workflow

1. Load the World Happiness Report Excel dataset into Google Colab.
2. Inspect the dataset structure, column names, data types and missing values.
3. Remove rows where the main Life Ladder measure is unavailable.
4. Group countries by average Life Ladder to identify the top 10.
5. Create a horizontal bar chart for the top 10 countries.
6. Create scatter plots for GDP, social support and freedom against Life Ladder.
7. Add clear titles, axis labels and one-line takeaways.
8. Connect all four charts into a single narrative.

9. Key Learning

This task demonstrates that effective data visualization is not simply about creating charts. Each visualization should answer a clear question, use an appropriate chart type, and contribute to an overall story. The project also provides practical experience with Pandas, Matplotlib, data cleaning, grouping, and visual interpretation in Google Colab.

10. Project Deliverables

- Google Colab / Jupyter Notebook with cell-wise Python code
- Four labeled visualizations
- One-line takeaway for each visualization
- Short narrative connecting the charts
- World Happiness Report dataset
- Visualization image files for documentation

11. Conclusion

The World Happiness Data Story converts country-level well-being data into a simple visual narrative. By starting with the highest life evaluations and then examining economic well-being, social support, and freedom, the project shows how multiple indicators can be explored together to better understand differences in reported happiness.

Internship: Veda Technologies – Data Analytics Internship

Task: Task 7 – First Chart Story: Turning Numbers into a Narrative

Visualization Tool: Python Matplotlib

Environment: Google Colab